

Paper Manager pack — Open X-Embodiment (Nov 2 · Data)

Paper: Open X-Embodiment: Robotic Learning Datasets and RT-X Models

arXiv: <https://arxiv.org/abs/2310.08864>

PDF: /workspace/papers/open-x-embodiment-2310.08864.pdf

Session: Nov 2 · due 9:30 AM · **You = Visionary (seat 1)** · Type: **Data**

Sources: Summarizer + Senior Research (/workspace/oxe-quiet/DELIVERABLE.md)

1. One-liner

OXE = municipal water for generalist robot policies; BEHAVIOR’26 = bottled specialty blend — steal mix science, don’t drink the river.

Shared multi-lab demo pool + RT-1-X/RT-2-X proving positive transfer without embodiment adapters — foundation for Octo/OpenVLA/ π_0 /BEHAVIOR data stories.

2. Exact pool vs train mix (CLEAR)

	Pool	RT-X train mix
Scale	**60** datasets / **22** embodiments / **1M+** traj	**9** manipulators ($\neq$ full 22)
Schema	RLDS	same
Eval	—	**3600** trials / **6** robots

Headline morals: Fig.4 ~**+50%** relative small-domain lift from co-training; Table I RT-1-X **underfits** large native Google set (73% vs RT-1 92%); Table II emergent **27.3% → 75.8%** (~3×); no-Bridge **42.8%**.

3. Keywords

4. Role packs

Stakeholder (18+3)

Open shared pool enabled community generalists without each lab building PaLI-scale data. Cite as infrastructure win; treat RT-2-X as transfer existence proof not production VLA; invest in modality coverage + long-horizon, not only more pick-place hours.

Scientific Reviewer (8+3)

Holds: matched RT-1 vs RT-1-X isolates mix; capacity caveat honest; Bridge ablation. Punish: “generalist X-robot” ≠ zero-shot new hardware; 9<<22 blur; PaLM skill ontology; 55B unrepro. Verdict: landmark systems+dataset paper — not complete BEHAVIOR recipe.

Empiricist (11+3)

Priority	Experiment
P0 Y	RLDS shard inventory/loader smoke; mix-cardinality (Bridge-only vs +2 vs +5) on open policy FT
P0 Partial	Leave-Bridge-out (Table II moral)
P1	Small-domain lift proxy; capacity×mix; embodiment leave-one-out; same mix → Octo/OpenVLA/ π_0
Defer	Full RT-1-X / RT-2-X retrain; TB-class full OXE mirror

Env: /workspace/oxe-quiet/env_outline/. Separate pool vs RT-X mix vs Octo 800k vs OpenVLA 970k.

Archaeologist (11+3)

RoboNet/lab demos/QT-Opt/Bridge/RT-1 → **OXE+RT-X** → Octo/OpenVLA soups → π_0 /OpenPI/ $\pi_{0.5}$ /GR00T → BEHAVIOR’25–’26. OXE replaced single-lab pretrain assumption; did **not** replace need for post-train demos, wrist/proprio, continuous chunks, challenge sim alignment.

Visionary (11+3) — ****YOU (seat 1)**** — HIGH PRIORITY

Steal from OXE

1. Embodiment/scene **width** as first-class
2. **Schema before scale** (RLDS → lock obs keys early)

3. Language as mix **column** (push BEHAVIOR post-train **>90%** + paraphrases)
4. Transfer **ablations** (Bridge-removal gold) + publish negative transfer
5. **Capacity × mix** co-design (Table I)

Anti-patterns → fixes

OXE-era fail	Fix for BEHAVIOR
Uniform everything	Explicit **weight table**
Loyalty to bad shards	**Kill switch** → weight 0
Wrist poverty	Hard gate contact tasks
Lang poverty	Dual-condition lang+subgoals
7×256 bins only	Chunked continuous (or FAST)
Huge shards drown gold	**Sublinear** size temperature ($\pi_0 n^{0.43}$)
Pick-place monopoly	Cap skill family $\leq \sim 20\%$

Concrete recipe (stages)

1. **Prior:** OXE-like / vendor $\pi_{0.5}$ /GR00T; sublinear weights; exclude bad shards
2. **Mid-train sim:** cover all tasks; upsample long-horizon + recovery; dense lang/subtask tokens
3. **Post-train gold:** highest sample weight; per-task floors; wrist-complete; aim BEHAVIOR tokens **≥30–50%** of late training
4. **Holdout mix science:** ablate no-prior / no-wrist / no-paraphrase / uniform vs $n^{0.43}$ / kill switch

Red lines: no uniform raw OXE dump; no “discrete bins enough for 2026”; no “OXE alone = BEHAVIOR plan.”

Skip Discord/Drive.

5. Slide Maker brief

Role	Timing	Focus
Stakeholder	18+3	Infrastructure win
Reviewer	8+3	Landmark + overclaim punish
Empiricist	11+3	Mix-cardinality smoke
Archaeologist	11+3	OXE → soups → BEHAVIOR
Visionary	**11+3**	**Mix cards + BEHAVIOR recipe (HIGH)**

GitHub title: [ECE 605] - Open X-Embodiment Making Visionary

Paths

Artifact	Path
This pack	`/workspace/oxe-quiet/KANTA_PACK.md`
Senior Research	`/workspace/oxe-quiet/DELIVERABLE.md`
Summarizer	`/workspace/papers/open-x-embodiment-2310.08864-summary.md`
Paper PDF	`/workspace/papers/open-x-embodiment-2310.08864.pdf`